

Signals & Systems (EE504)
Pre-Reading Material – Lecture 1

Signals & Systems
Estimated Reading Time: 20–25 minutes

Course Context

This document provides an introduction to **EE 504 – Signals and Systems**. The course introduces the fundamental concepts used to represent, analyze, and design systems that process signals. It forms the foundation for advanced subjects such as Digital Signal Processing, Communication Systems, Control Systems, Robotics, and Machine Learning.

Recommended Textbooks

- A. V. Oppenheim, A. S. Willsky, and I. T. Young, *Signals and Systems*.
- G. R. Cooper and C. D. McGillem, *Methods of Signal and System Analysis* (for Random Signals and Systems).

Why This Topic Matters

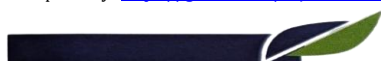
Signals and systems are fundamental to almost every engineering application. Whether in communication, biomedical engineering, electronics, or automation, engineers analyze signals and design systems that transform information into useful outputs.

For example, in an ECG monitoring system, the electrical activity of the heart (input signal) is processed by a digital system to assist doctors in diagnosing heart conditions.

Learning Outcomes

After studying this lecture, you should be able to:

- Define signals and systems.
- Distinguish between analysis and synthesis problems.
- Differentiate between continuous-time and discrete-time signals.
- Understand the basic idea of sampling and its importance in digital systems.



Prior Knowledge

Recall your laboratory experiment on an RC circuit. The circuit receives an input voltage and produces an output voltage by performing a specific electrical operation. This is a simple example of a system transforming one signal into another.

Key Concepts

Signal

A **signal** is a physical quantity that varies with one or more independent variables and carries information. In this course, signals are primarily studied as functions of time.

System

A **system** is a meaningful interconnection of components that performs a specific function by processing an input signal to produce an output signal.

Analysis

Analysis refers to determining the output of a system when both the input signal and the system are known.

Synthesis

Synthesis (or design) refers to developing a system that produces a desired output for a given input. Unlike analysis, a synthesis problem may not always have a feasible solution.

Continuous-Time Signal

A **continuous-time (CT) signal** is defined at every instant of time.

Discrete-Time Signal

A **discrete-time (DT) signal** is defined only at specific time instants, usually obtained through sampling.



Sampling

Sampling is the process of converting a continuous-time signal into a discrete-time signal by recording its values at regular intervals.

Guided Reading Questions

Before attending the lecture, think about the following:

1. What is the difference between a collection of components and a system?
2. Why are many real-world signals converted into discrete-time signals?
3. Why might a synthesis problem not always have a solution?

Self-Assessment

True or False

A synthesis problem always has a solution.

Answer: False.

Short Answer

What is the difference between a continuous-time signal and a discrete-time signal?

Answer: A continuous-time signal exists at every instant of time, whereas a discrete-time signal exists only at specific sampling instants.

